CONFIGER IPv6 ADRESSING

1.Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).

ANS:

Practical: Assign and Verify a Static IPv6 Address

Aim: To assign a static IPv6 address to the primary network adapter and verify the configuration using command-line tools.

Method 1: Windows

Step 1: Open Command Prompt as Administrator.

Step 2: Identify the network adapter.

netsh interface ipv6 show interfaces

Step 3: Assign a static IPv6 address.

Example using an adapter named Ethernet:

netsh interface ipv6 set address interface="Ethernet" address=2001:db8:1::10

2001:db8:1::10 is an example address for documentation and is not intended for real Internet connectivity.

Step 4: Verify the configuration.

ipconfig

Look for the IPv6 Address under the correct adapter.

Method 2: Linux

Step 1: Identify the network adapter.

ip link show

Step 2: Assign a static IPv6 address.

Example using interface eth0:

sudo ip -6 addr add 2001:db8:1::10/64 dev eth0

Replace eth0 with your actual adapter name.

Step 3: Verify the configuration.

ip -6 addr show dev eth0

Expected Result

Item	**Example**

Network Adapter	Ethernet / eth0
IPv6 Address	2001:db8:1::10
Prefix Length	/64
Verification Command	ipconfig or ip -6 addr show
Result	IPv6 address appears under the selected adapter

**2.Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address.

Hint: Use the ping6 or ping -6 command for IPv6 testing.**

ANS:

Practical: Configure and Test a Unique Local IPv6 Address

Aim: To assign a Unique Local IPv6 Address (ULA) starting with fd and test connectivity with another device on the same LAN.

Step 1: Assign an IPv6 address

For Windows (Command Prompt as Administrator):

First, check your adapter name:

netsh interface ipv6 show interfaces

Assign an example address to your Ethernet adapter:

netsh interface ipv6 set address interface="Ethernet" address=fd12:3456:789a:1::10

For Linux:

sudo ip -6 addr add fd12:3456:789a:1::10/64 dev eth0

Replace eth0 with your actual network interface name.

Step 2: Configure the second device

Assign another unique address in the same /64 subnet to the second device:

fd12:3456:789a:1::20/64

Both devices must be connected to the same LAN, and their firewall settings must allow IPv6 ping requests.

Step 3: Verify the IPv6 address

Windows:

ipconfig

Linux:

ip -6 addr show

Check that the assigned address appears on the correct adapter.

Step 4: Test connectivity

From the first device, ping the second device.

Windows:

ping -6 fd12:3456:789a:1::20

Linux:

ping -6 fd12:3456:789a:1::20

Expected Result

Test	**Expected Output**
IPv6 configuration	Address beginning with fd appears
Ping test	Replies received from the second device
Result	Successful IPv6 communication on the LAN

**3.Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect.

Constraint: Document the steps and the observed address behavior in a text file.**

ANS:

PRACTICAL: TEMPORARY IPv6 ADDRESS — TEST AND OBSERVATION REPORT

Aim To configure a temporary IPv6 address and check whether it changes after a reboot or after disconnecting and reconnecting the network.

Important I cannot access or reboot your computer from this chat, so the actual observations must be recorded on your own system. Do not report a change unless you verify it.

A. Windows steps 1. Open Command Prompt. 2. Identify the network adapter: ipconfig /all 3. View IPv6 addresses: ipconfig 4. For a temporary test address, open Command Prompt as Administrator and add an address to the correct adapter (replace “Ethernet” with your adapter name): netsh interface ipv6 add address interface=“Ethernet” address=fd12:3456:789a:1::50 This is a Unique Local IPv6 address for a lab test; it is not a globally routable Internet address. 5. Verify: ipconfig 6. Remove the test address after the experiment (use the same adapter and address): netsh interface ipv6 delete address interface=“Ethernet” address=fd12:3456:789a:1::50

B. Linux steps 1. Identify the network interface: ip link show 2. View current IPv6 addresses: ip -6 addr show 3. Add a temporary lab address (replace eth0 with your actual interface): sudo ip -6 addr add fd12:3456:789a:1::50/64 dev eth0 4. Verify: ip -6 addr show dev eth0 5. Remove the test address after the experiment: sudo ip -6 addr del fd12:3456:789a:1::50/64 dev eth0

C. Test after network reconnect 1. Record the IPv6 address before disconnecting. 2. Disconnect and reconnect the network adapter (or Wi-Fi). 3. Run ipconfig on Windows or ip -6 addr show on Linux. 4. Record whether the address remains or disappears/changes.

D. Test after reboot 1. Save your work and reboot the computer. 2. Run the same verification command. 3. Record whether the test address remains or disappears/changes.

Observation table (fill in after performing the test) - Operating system: _____ - Adapter/interface name: _____ - Address before reconnect: _____ - Address after reconnect: _____ - Reconnect result (same / changed / disappeared): _____ - Address after reboot: _____ - Reboot result (same / changed / disappeared): _____ Expected behavior / explanation The example address above is manually added for testing. On many systems, an address added with these commands is not persistent and may disappear after reboot; behavior after disconnect/reconnect can vary by operating system and interface. IPv6 privacy addresses generated automatically by the operating system are different: they may rotate or change over time depending on system and network settings. Record the actual result from your computer rather than assuming the address changed.

4.Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.

ANS:

Python Script: Zomato-Style Restaurant Search Using IPv6

This program stores restaurant names and their IPv6 addresses in an array (list). The user enters a partial restaurant name, and the program displays matching restaurants with their IPv6 addresses.

Zomato-Style Restaurant Searchrestaurants = [{"name": "Pizza Palace", "ipv6": "fd12:3456::1"}, {"name": "Burger House", "ipv6": "fd12:3456::2"}, {"name": "Spice Garden", "ipv6": "fd12:3456::3"}, {"name": "Royal Biryani", "ipv6": "fd12:3456::4"}]search = input("Enter restaurant name: ").lower()found = Falsefor restaurant in restaurants: if search in restaurant["name"].lower(): print("Restaurant:", restaurant["name"]) print("IPv6 Address:", restaurant["ipv6"]) found = Trueif not found: print("No matching restaurant found.")

Sample Output

Enter restaurant name: pizza

Restaurant: Pizza Palace

IPv6 Address: fd12:3456::1

Explanation

1. The list stores restaurant names and their IPv6 addresses.

2. The user enters a full or partial restaurant name.
3. The program searches names without considering uppercase or lowercase letters.
4. Matching restaurant names and IPv6 addresses are displayed.
5. If no restaurant matches, an error message is shown.

5.Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

ANS:

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IPv6 Subnetting Scenario and Configuration Steps

1. AI-Generated Scenario

A college wants to create three separate IPv6 subnets for the Computer Lab, Library, and Administration Department using the network prefix fd12:3456:789a::/48.

Department	Network Prefix	Example IPv6 Address
Computer Lab	fd12:3456:789a:1::/64	fd12:3456:789a:1::10
Library	fd12:3456:789a:2::/64	fd12:3456:789a:2::10
Administration	fd12:3456:789a:3::/64	fd12:3456:789a:3::10

Each department has a separate /64 IPv6 subnet. These addresses are Unique Local IPv6 Addresses (ULA) used for private network communication.

2. Configuration Steps for Windows

Step 1: Open Command Prompt as Administrator.

Step 2: Identify the network adapter.

netsh interface ipv6 show interfaces

Step 3: Assign an IPv6 address to the Ethernet adapter.

netsh interface ipv6 add address interface="Ethernet" address=fd12:3456:789a:1::10

Step 4: Verify the IPv6 configuration.

ipconfig

Check whether the assigned IPv6 address appears under the Ethernet adapter.

3. Linux Configuration Commands

Step 1: Identify the network interface.

ip link show

Step 2: Assign an IPv6 address.

sudo ip -6 addr add fd12:3456:789a:1::10/64 dev eth0

Step 3: Verify the address.

ip -6 addr show

Note: Replace eth0 or Ethernet with the actual interface name on your computer.